Hospital Readmission Analysis - Findings & Methodology

Final write-up for the SQL/BI hospital readmission reporting portfolio project
Diabetes 130-US Hospitals dataset (1999-2008)
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Overview

This project analyzes 101,766 diabetic inpatient encounters across 130 US hospitals and integrated delivery networks (1999-2008), sourced from the UCI Diabetes 130-US hospitals dataset (Strack et al., 2014). While many analyses of this dataset include building a machine learning classifier to predict readmission, this project was deliberately scoped as a SQL- and BI-first reporting pipeline: raw CSV -> Python ETL -> normalized MySQL schema -> SQL views -> SQL reporting queries -> a four-page Power BI dashboard. The goal was to mirror how a hospital quality or operations team would consume this kind of data: through transparent, auditable queries and a dashboard.

After excluding 2,423 terminal-discharge encounters (death and hospice dispositions, 2.38% of the raw data), the reportable base for all analysis is 99,340 encounters, with an overall 30-day readmission rate of 11.39%.

Findings

1. A simple, transparent risk score can effectively separate readmission risk

A three-factor, rule-based risk score using prior-year hospital use, high-readmission diagnosis categories, and non-home discharge separated encounters into Low, Medium, and High risk groups. Readmission rates increased from 7.90% (Low) -> 13.93% (Medium) -> 18.69%(High), with the High-risk group having a 2.37x higher readmission rate than the Low-risk group. High-risk encounters made up 16.54% of all encounters.

Since the scoring rules are directly defined in SQL, they are easy to explain, audit, and change as needed.

2. Prior-year utilization is a stronger indicator of readmission risk than the combined risk score

When prior-year utilization (outpatient + emergency + inpatient visits) is examined on its own, bucketed into 0, 1-2, 3-5, and 6+ visit groups, readmission rate increases. Encounters with 0 prior visits had an 8.3% readmission rate, compared with 12.9% for 1-2 visits, 16.9% for 3-5 visits, and 26.0% for 9 or more visits. This represents a 3.13x spread between the lowest and highest utilization groups, compared with a 2.37x difference across the composite risk tiers.

Prior utilization is doing most of the predictive work in the composite score. Hospitals could use this readily available information to help identify higher-risk patients, even before considering diagnosis and discharge details.

3. Discharge destination is strongly associated with readmission rates

Diagnosis and clinical severity are often treated as the primary drivers of readmission risk, with discharge logistics treated as secondary. Among the ten most common discharge dispositions (covering the large majority of all encounters), readmission rates ranged from 9.30% for patients discharged home to 36.69% for patients discharged to a psychiatric facility. This is a near 4x difference, the widest of any factor examined in this project.

Where a patient goes after discharge may be an important indicator of readmission risk, alongside their underlying diagnosis. This points to post-discharge care coordination and transition planning, not just clinical severity, as a high-leverage area for intervention.

4. Diagnosis category shows a real, if modest, readmission penalty signal

It's reasonable to expect that some diagnoses are simply "harder" cases with mechanically higher readmission risk, and to assume this effect in large. Using an internal baseline-proxy ratio (each category's own readmission rate divided by the dataset's overall rate), Diabetes over-indexes at 1.15x baseline (13.10% category rate), while Musculoskeletal under-indexes at 0.84x (9.54%). Although the diagnosis category showed a consistent difference in readmission rates, it had the smallest effect among the five findings in this report.

Diagnosis-specific protocol is justifiable but should be weighted modestly relative to utilization history and discharge planning. This proxy is intentionally calculated from within the dataset rather than using an official CMS Hospital Readmission Reduction Program (HRRP) Excess Readmission Ratio, which needs facility-level data not available here.

5. Medication increases show a small association with readmission, not proof of cause

Encounters with a medication dosage increase had a readmission rate of 12.94%, compared to 11.15% for encounters without an increase. Although the gap is real, it is small (under 2%), making it the smallest readmission-rate gap among the five findings.

Medication increases may reflect that a patient was more seriously ill and needed treatment adjustments, rather than causing a higher risk of readmission. It's worth tracking as a minor contextual flag, but shouldn't be treated as a strong standalone risk indicator. The results show an association, not cause and effect.

Methodology & Limitations

Scope and design decisions

This project's defining choice was to build a SQL/BI reporting pipeline instead of a predictive model, as a deliberate differentiation strategy. Every finding in this report is derived from auditable SQL views and reporting queries, not a trained model.

Another deliberate decision made was not to replicate the original Strack et al. (2014) paper's restriction to one encounter per patient. In the source paper, the logistic's regression model required statistically independent observations; hence the restriction. In this project, all 99,340 reportable encounters were kept to show how a team would count and track readmissions per encounter rather than per patient.

Data Limitations

- No date or timestamp field exists in the source data. ‘encounter_id’ is used as a chronological proxy where the order matters.
- The penalty proxy ratio is explicitly internal-only. It compares each diagnosis category’s readmission rate to this dataset’s own overall average. It isn’t, and isn’t presented as, an official CMS HRRP Excess Readmission Ratio, which would require facility-level claims data that this dataset does not have.
- Several categories rest on very small sample sizes and are excluded from ranked or rate-based charts. Specifically, the “Unknown” diagnosis category (20 encounters) and “Other” (a non-clinical catch-all). Both are kept in the underlying data and views, but excluded from visualizations where their small size would distort a ranking.
- Age group is modeled at the encounter grain, not the patient grain. Profiling found that 1,574 patients had more than one distinct age value across encounters. Of those, 98.6% showed genuine aging across the 1999-2008 span. This confirmed that age is a time-varying attribute for this dataset and shaped the final schema design.

Conclusion

The strongest readmission signals in this dataset were related to a patient’s prior-year utilization and discharge destination, rather than their diagnosis. Prior utilization showed a 3.13x difference in readmission rates, while discharge destination showed a nearly 4x difference. In comparison, diagnosis category (1.15x down to 0.84x against baseline) and medication increases (12.94% vs. 11.15%) had smaller differences. Both strong signals come from administrative fields that almost every hospital already records, and all of these are associations, not causal effects.

Practical takeaways

- Flag patients with three or more prior-year visits for early follow-up planning. The 3-5 and 6+ visit groups readmit at 16.9% and 26.0%.
- Prioritize transition planning for non-home discharges, especially to psychiatric and rehab facilities (36.69% and 27.71%).
- Treat diagnosis category and medication changes as context in a risk review, not as standalone triggers.

Future work

- Real admission dates would allow trend analysis and true time-to-readmission measures, which this dataset cannot support.
- Facility-level data would allow a genuine HRRP Excess Readmission Ratio in place of the internal proxy used here.
- Benchmarking the rule-based risk tier against a simple predictive baseline would show how much, if anything, a model adds over a transparent score.